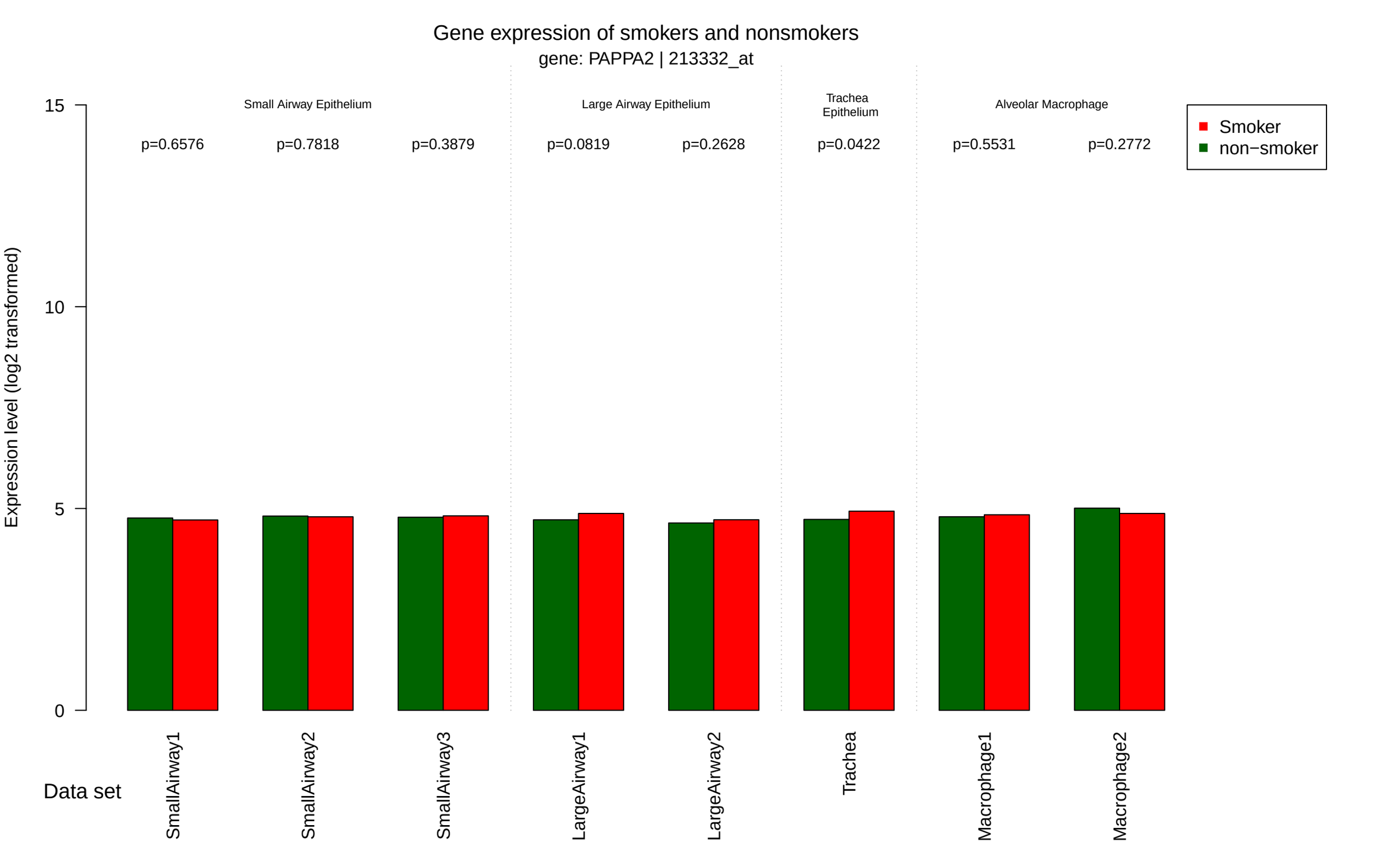
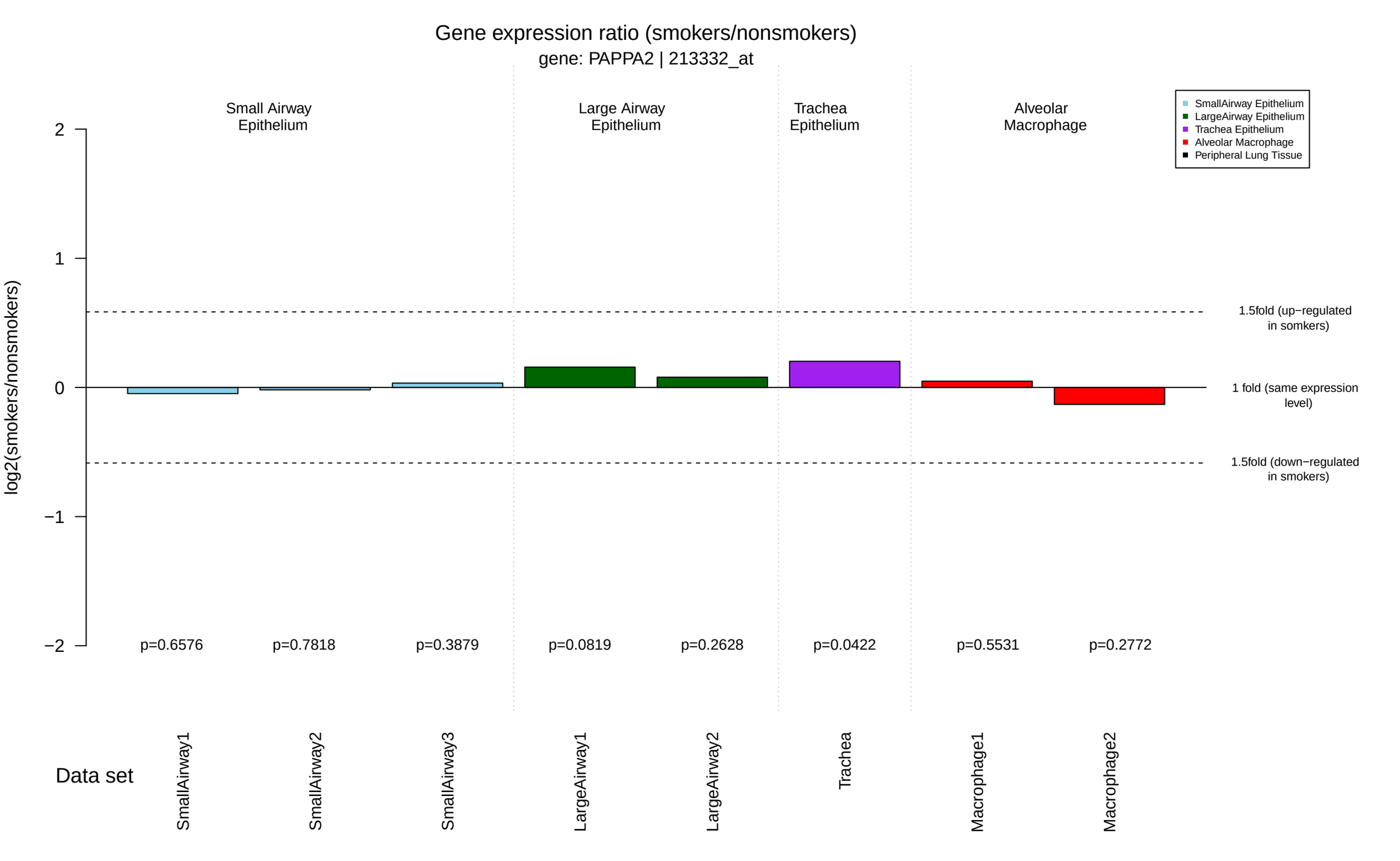
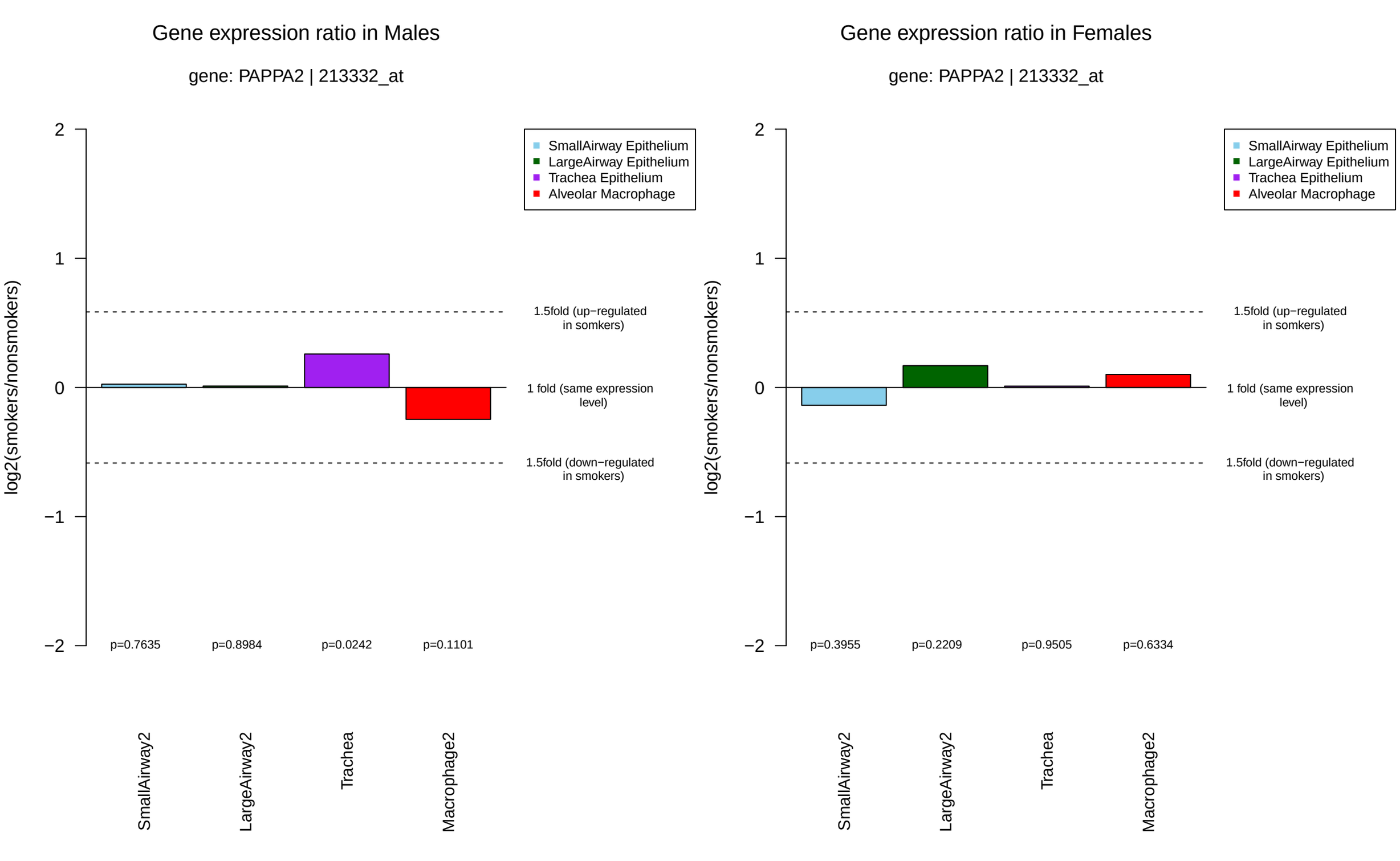
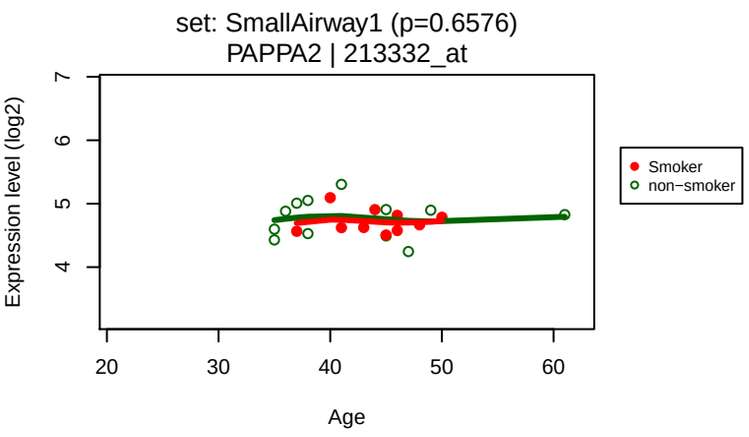
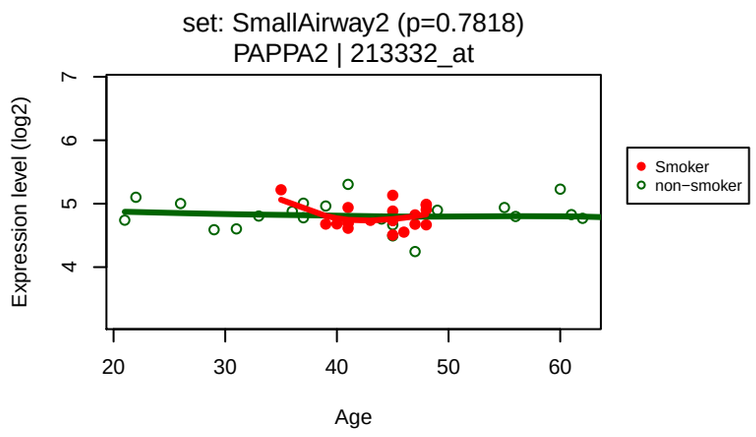
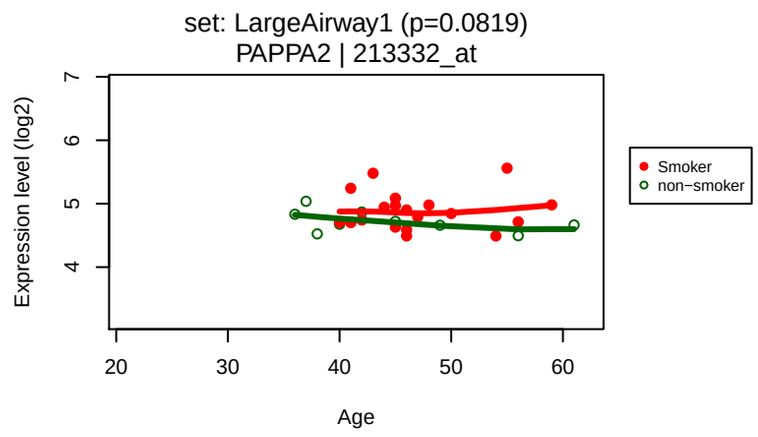

Gene expression of smokers and nonsmokers
gene: PAPPA2 | 213332_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6576
p=0.7818
p=0.3879
p=0.0819
p=0.2628
p=0.0422
p=0.5531
p=0.2772
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

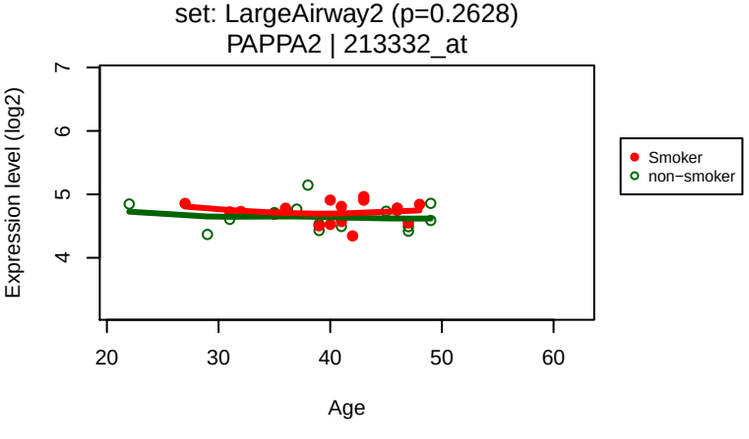

Gene expression ratio (smokers/nonsmokers)
gene: PAPPA2 | 213332_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6576
p=0.7818
p=0.3879
p=0.0819
p=0.2628
p=0.0422
p=0.5531
p=0.2772
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

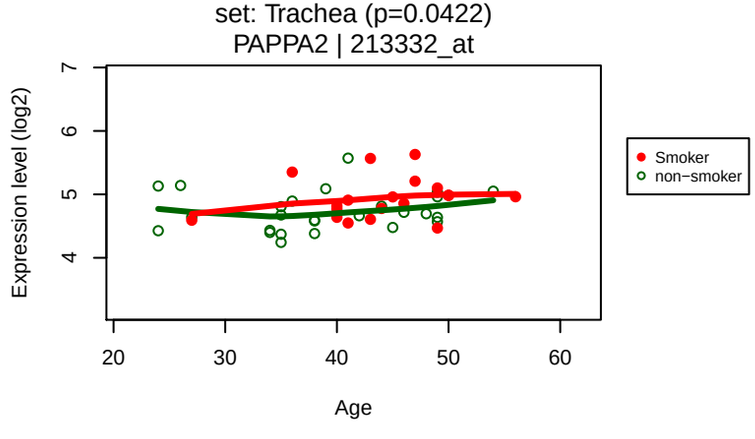

Gene expression ratio in Males
gene: PAPPA2 | 213332_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7635
p=0.8984
p=0.0242
p=0.1101
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAPPA2 | 213332_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3955
p=0.2209
p=0.9505
p=0.6334
SmallAirway2
LargeAirway2
Trachea
Macrophage2

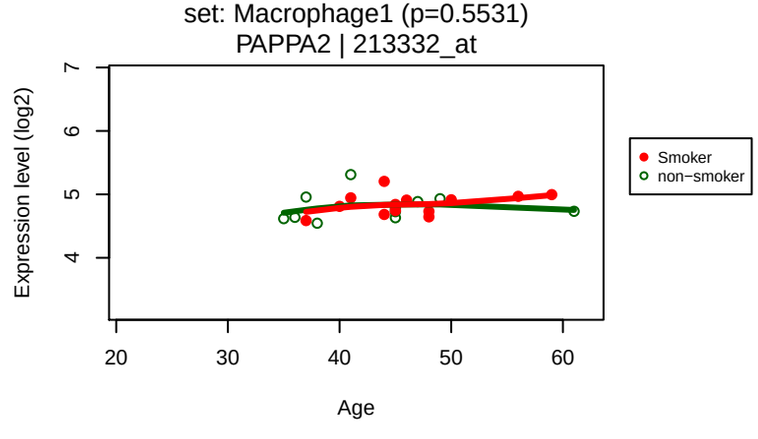

set: SmallAirway1 (p=0.6576)
PAPPA2 | 213332_at
Expression level (log2)
Age
Smoker
non−smoker

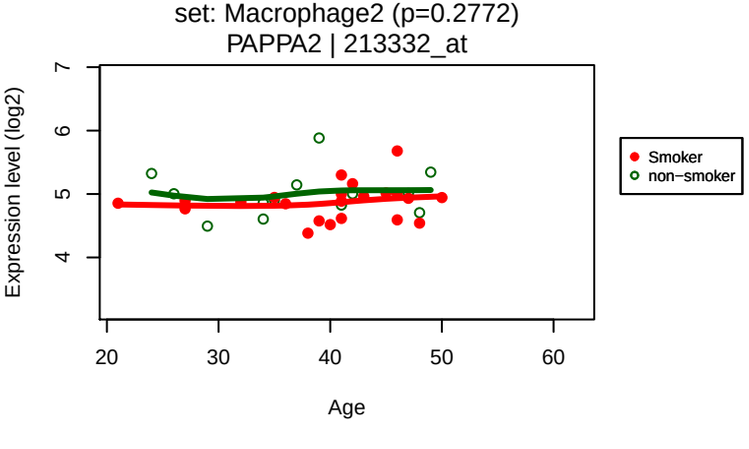

set: SmallAirway2 (p=0.7818)
PAPPA2 | 213332_at
Expression level (log2)
Age
Smoker
non−smoker

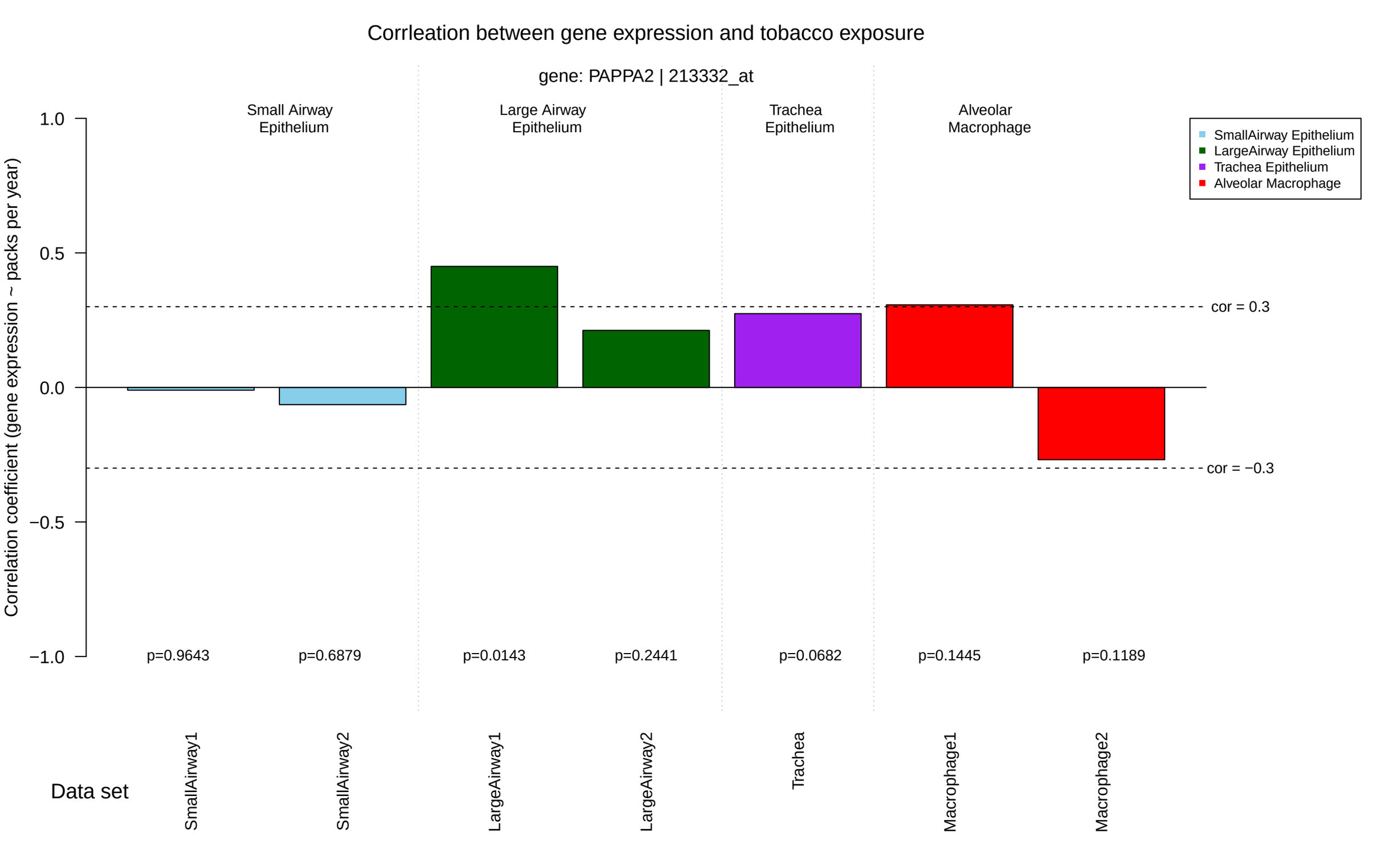

set: LargeAirway1 (p=0.0819)
PAPPA2 | 213332_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2628)
PAPPA2 | 213332_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0422)
PAPPA2 | 213332_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5531)
PAPPA2 | 213332_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2772)
PAPPA2 | 213332_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAPPA2 | 213332_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9643
p=0.6879
p=0.0143
p=0.2441
p=0.0682
p=0.1445
p=0.1189
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2